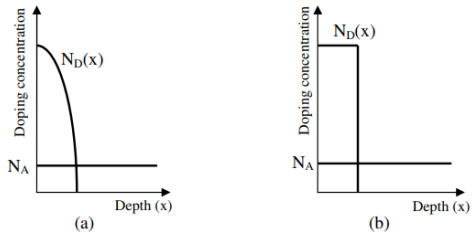
Solid State Devices EE3001 Quiz 2 Syllabus

Simple p-n junction

Physically, this means that the p-region has a large concentration
of holes (and few electrons) while n-region has a large concentration of electrons (and few holes). Now, when a p-region and an n-region are brought in intimate contact (i.e. a junction is formed), this large concentration gradient at the junction causes diffusion of carriers. While holes diffuse from the p-side to the n-side, electrons diffuse from the n-side to the p-side.

Consequently, a negative space charge builds up on the p-side and a positive space charge on the n-side. This creates a built-in electric field
directed from positive charge to negative charge (i.e. from n-side to p-side) which gives rise to a drift current. The direction of this drift current will be opposite to that of the diffusion current for both electrons and holes as shown in Fig. 4.2(b). An equilibrium condition is reached where there is no net transport of carriers as the diffusion component of current is exactly balanced by an equal and opposite drift component.



4.1(a) Actual doping profile in a p-n junction and (b) the approximate abrupt doping profile

The difference in E_c/E_v from the p-side to the n-side is given by qV_{bi}, where V_{bi} is the called the contact potential or the built-in potential.

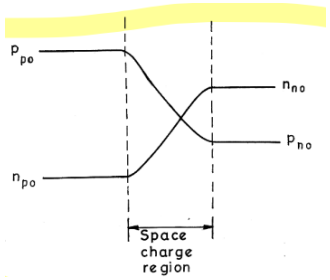
The built-in potential cannot be measured by simply connecting a voltmeter across the junction as the sum of the contact potentials between metal and semiconductor exactly compensate the built-in potential of the p-n junction.

$$E_{ip} - E_{in} = kT \ln\left(\frac{N_A N_D}{n_i^2}\right) = qV_{bi}$$

$$\text{Assuming complete ionization: } p_{p0} \approx N_A \text{ and } n_{n0} \approx N_D.$$

$$V_{bi} = V_T \ln\left(\frac{p_{p0}}{p_{n0}}\right) = V_T \ln\left(\frac{n_{n0}}{n_{p0}}\right)$$

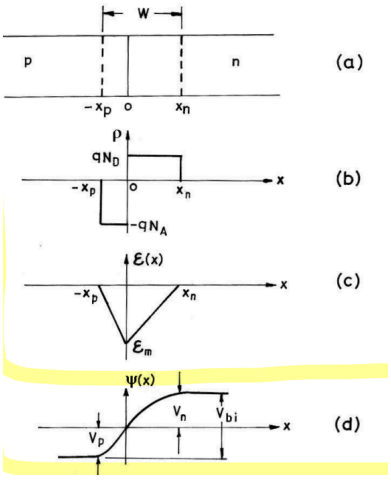
So, we can consider that within the space charge region the charge is only due to the immobile acceptor and donor ions. In other words, this region is depleted of mobile carriers and is also referred to as depletion layer. In this
depletion region, as we move from the p-side to n-side, the difference between the intrinsic level and Fermi level at first reduces, becomes zero and then becomes negative



we assume that the space charge region has sharp boundaries, where the charge concentration changes abruptly from $\rho_p = -qN_A$ to zero on the p-side and $\rho_n = qN_D$ to zero on the n-side.

the total uncompensated charge on each side of the junction is given by $Q_D = AqN_Ax_p = AqN_Dx_n$

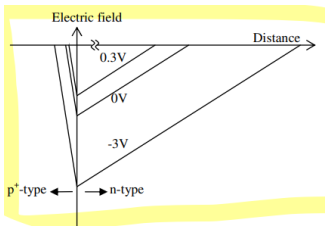
— 1. Poisson Equation for Depletion Region —
Assuming a step junction with complete ionization:
$\frac{d\mathcal{E}}{dx} = \frac{q(N_D - N_A)}{\epsilon_s}$
Specifically:
$\frac{d\mathcal{E}}{dx} = \frac{qN_D}{\epsilon_s}$ for $0 < x < x_n$
$\frac{d\mathcal{E}}{dx} = -\frac{qN_A}{\epsilon_s}$ for $-x_p < x < 0$
— 2. Electric Field Distributions —
Integrating the equations above with boundary conditions $\mathcal{E}(x_n) = 0$ and $\mathcal{E}(-x_p) = 0$:
$\mathcal{E}_p(x) = -\frac{qN_A(x+x_p)}{\epsilon_s} = \mathcal{E}_m\left(1 + \frac{x}{x_p}\right)$ for $-x_p < x < 0$
$\mathcal{E}_n(x) = \frac{qN_D(x-x_n)}{\epsilon_s} = \mathcal{E}_m\left(1 - \frac{x}{x_n}\right)$ for $0 < x < x_n$
— 3. Peak Electric Field ($\mathcal{E}_m$) —
The maximum field occurs at the metallurgical junction (x = 0):
$\mathcal{E}_m = -\frac{qN_Dx_n}{\epsilon_s} = -\frac{qN_Ax_p}{\epsilon_s}$
— 4. Potential Drop as Area of the Field Triangle —
The built-in potential is the negative integral of the electric field profile:
$V_{bi} = -\int_{-x_p}^{x_n} \mathcal{E}dx = -\int_{-x_p}^{0} \mathcal{E}_p dx - \int_0^{x_n} \mathcal{E}_n dx$
Geometrically, this is the area of the triangle formed by the $\mathcal{E}(x)$ plot:
$V_{bi} = \frac{q(N_Ax_p^2 + N_Dx_n^2)}{2\epsilon_s} = \frac{|\mathcal{E}_m|W}{2}$
— 5. Final Depletion Width (W) —
Solving for the total depletion width $W = x_p + x_n$:
$W = \sqrt{\frac{2\epsilon_s V_{bi}}{q}\left(\frac{1}{N_A} + \frac{1}{N_D}\right)} = \sqrt{\frac{2\epsilon_s V_{bi}}{qN_{eff}}}$
where $N_{eff} = \frac{N_AN_D}{N_A+N_D}$ is the effective doping concentration.

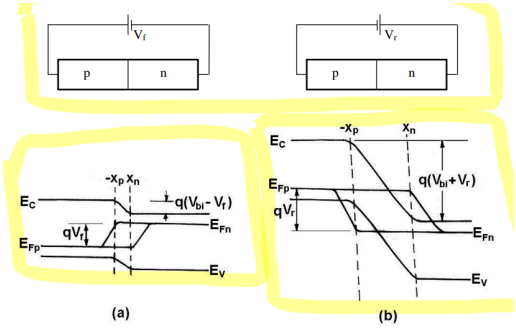


p-n junction under applied bias

When the p-n junction is forward biased by V_f, i.e. the positive terminal of the battery is connected to the p-side and the negative terminal to the n-side, the potential difference across the space charge region is reduced to $V_{bi} - V_f$. On the other hand, when the p-n junction is reverse biased by V_r, i.e. the positive terminal of the battery is connected to the n-side and the negative terminal to the p-side, the potential difference is increased to $V_{bi} - V_r$

$V_{bi} - V = \frac{q(N_Ax_p^2 + N_Dx_n^2)}{2\epsilon_s} = \frac{|\mathcal{E}_m|W}{2}$; $W = \sqrt{\frac{2\epsilon_s(V_{bi} - V)}{qN_{eff}}}$

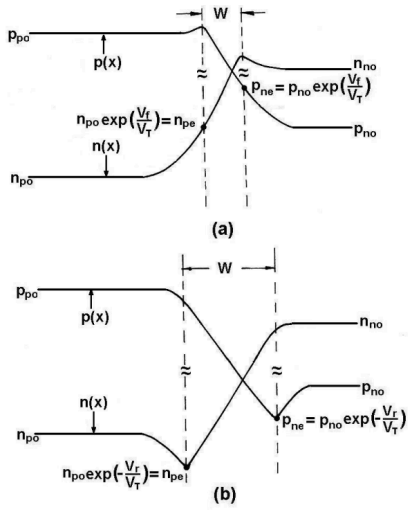




$$C_j = \frac{dQ_D}{dV} = \frac{A\epsilon_s}{W} = A\sqrt{\frac{q\epsilon_s N_D}{2(V_{bi}-V)}} = C_{j0}\left(1 - \frac{V}{V_{bi}}\right)^{-1/2}$$

Infinitely Long Diode

— 1. Equilibrium of Drift and Diffusion Currents —
In the depletion region, the hole drift and diffusion currents nearly cancel out:
$qp\mu_p\mathcal{E} \approx qD_p\frac{dp}{dx}$
— 2. Relation to Electrostatic Potential —
Using the Einstein relation ($D/\mu = V_T$) and $\mathcal{E} = -d\psi/dx$:
$\frac{d\psi}{dx} = -\frac{V_T}{p}\frac{dp}{dx}$
— 3. Integration Across the Depletion Region —
Integrating from the p-side edge (potential 0, conc. p_{pe}) to the n-side edge (potential $V_{bi} - V$, conc. p_{ne}):
$\int_0^{V_{bi}-V} d\psi = -V_T\int_{p_{pe}}^{p_{ne}}\frac{dp}{p}$
— 4. Carrier Concentration at the Depletion Edge —
Solving the integral and noting that $p_{pe} \approx p_{p0}$ (under low-level injection):
$p_{ne} = \frac{p_{pe}p_{n0}}{p_{p0}}\exp\left(\frac{V}{V_T}\right)$
— 5. The Law of the Junction (Injection Equations) —
The injected minority carrier concentrations are exponentially dependent on applied bias:
$p_{ne} = p_{n0}\exp\left(\frac{V}{V_T}\right)$
$n_{pe} = n_{p0}\exp\left(\frac{V}{V_T}\right)$



— 1. Minority Carrier Excess Profiles —
The injected carriers decay exponentially as they diffuse into the neutral regions:

$$p_n - p_{n0} = p_{n0}\left[\exp\left(\frac{V}{V_T}\right) - 1\right]\exp\left(-\frac{x - x_n}{L_p}\right)$$

$$n_p - n_{p0} = n_{p0}\left[\exp\left(\frac{V}{V_T}\right) - 1\right]\exp\left(\frac{x + x_p}{L_n}\right)$$

— 2. Diffusion Currents at the Depletion Edges —
Taking the derivative of the profiles to find diffusion current at the boundaries:

$$I_p(x_n) = -AqD_p \left.\frac{dp_n}{dx}\right|_{x=x_n} = \frac{AqD_p p_{n0}}{L_p}\left[\exp\left(\frac{V}{V_T}\right) - 1\right]$$

$$I_n(-x_p) = AqD_n \left.\frac{dn_p}{dx}\right|_{x=-x_p} = \frac{AqD_n n_{p0}}{L_n}\left[\exp\left(\frac{V}{V_T}\right) - 1\right]$$

— 3. Total Ideal Diode Current —
The total current is the sum of hole and electron currents at the edges:

$$I = I_p(x_n) + I_n(-x_p) = Aq\left(\frac{D_p p_{n0}}{L_p} + \frac{D_n n_{p0}}{L_n}\right)\left[\exp\left(\frac{V}{V_T}\right) - 1\right]$$

— 4. The Shockley Equation —

$$I = I_0\left[\exp\left(\frac{V}{V_T}\right) - 1\right]$$

— 5. Reverse Saturation Current (I_0) —

$$I_0 = Aq\left(\frac{D_p p_{n0}}{L_p} + \frac{D_n n_{p0}}{L_n}\right) = Aqn_i^2\left(\frac{D_p}{L_p N_D} + \frac{D_n}{L_n N_A}\right)$$

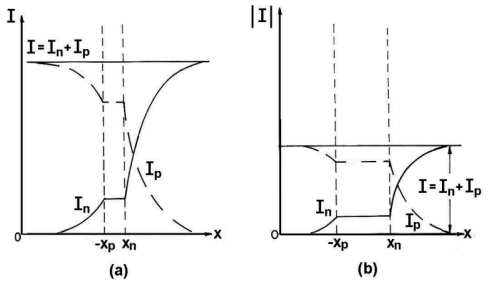
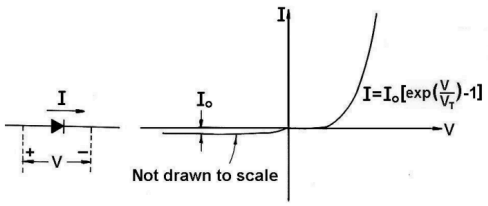


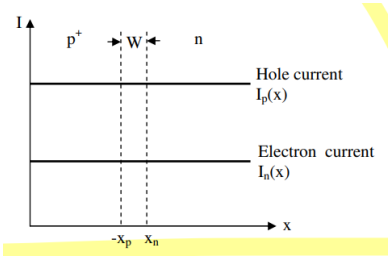
Fig. 4.11 Variation of electron and hole current components in a p-n junction under (a) forward bias and (b) reverse bias conditions



At edge of depletion region: $np = n_i^2 e^{\frac{V}{V_T}}$

Short Diodes

$I = Aqn_i^2 \left[\frac{D_p}{N_D W_n} + \frac{D_n}{N_A W_p}\right]\left[\exp\left(\frac{V}{V_T}\right) - 1\right] = I_0\left[\exp\left(\frac{V}{V_T}\right) - 1\right]$



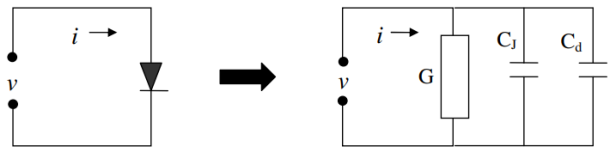
Time Variation of Stored Charge

— 1. Charge Control Equation —
The total current is the sum of the recombination rate and the rate of change of stored charge:
$I(t) = \frac{Q_p(t)}{\tau_p} + \frac{\partial Q_p(t)}{\partial t}$
— 2. Definition of Stored Excess Charge —
The total excess minority carrier charge stored in the neutral n-region is:
$Q_p = Aq\int_{x_n}^{\infty}(p_n - p_{n0})dx$
— 3. Charge Decay during Reverse Recovery —
When the diode is switched off, the stored charge decays exponentially over time:
$Q_p(t) = I\tau_p \exp\left(-\frac{t}{\tau_p}\right)$
— 4. Transient Voltage Response —
The voltage across the diode as a function of time during the turn-off transient is:
$V(t) = V_T \ln\left[\frac{I\tau_p}{AqL_p p_{n0}} \exp\left(-\frac{t}{\tau_p}\right) + 1\right]$

Charge Storage Capacitance

$G = \frac{dI}{dV} = \frac{I}{V_T}; C_d = G\tau_p$

— 1. Current in a Short Base Diode —
$I \approx I_p(x_n) = \frac{Q_p}{\tau_t}$
— 2. Transit Time Definition —
$\tau_t = \frac{W_n^2}{2D_p}$
— 3. Diffusion Capacitance (Short Base) —
$C_d = \frac{I\tau_t}{V_T} = G\tau_t$



$G = I/V_T$, $C_j = A\sqrt{\frac{q\epsilon_{Si}N_D}{2(V_{bi}-V)}}$ and $C_d = G\tau_t$

Breakdown Mechanisms

Example: In a p^+nn^+ junction diode, the doping concentration of the n-region is $2 \times 10^{15}\,\text{cm}^{-3}$. If the critical fie

(a) The breakdown voltage if the width of the n-region is $10\,\mu$m

(b) The breakdown voltage if the width of the n-region is $1\,\mu$m

Solution:

(a) For this structure, the depletion region exists essentially only in the n-region.

At breakdown the peak electric field is $|\mathcal{E}_m| = \mathcal{E}_c = 2 \times 10^5\,\text{V/cm}$.

Substituting this value and assuming $x_n = W$, we get:

$$W = \frac{2 \times 10^5 \times 11.9 \times 8.854 \times 10^{-14}}{1.6 \times 10^{-19} \times 2 \times 10^{15}}\,\text{cm} = 6.585\,\mu\text{m}.$$

The breakdown voltage is now obtained as the area of the triangle:

$$V_{BR} = \frac{2 \times 10^5 \times 6.585 \times 10^{-4}}{2} = 65.85\,\text{V}$$

(b) Since the width of the n-region is now $1\,\mu$m, the depletion region width is restricted to $1\,\mu$m.

The slope of the electric field distribution in the n-region is:

$$\frac{d\mathcal{E}}{dx} = \frac{qN_D}{\epsilon_s} = \frac{1.6 \times 10^{-19} \times 2 \times 10^{15}}{11.9 \times 8.854 \times 10^{-14}} = 3.037 \times 10^8\,\text{V/cm}^2$$

The electric field takes a trapezoidal shape. We calculate the field at the nn^+ junction:

$$\mathcal{E}_A = 2 \times 10^5 - (3.037 \times 10^8 \times 10^{-4}) = 1.696 \times 10^5\,\text{V/cm}$$

The breakdown voltage is the area under the trapezoid:

$$V_{BR} = \frac{(2 + 1.696) \times 10^5}{2} \times 10^{-4} = 18.48\,\text{V}$$

